

## EC466: Econometrics

### Problem Set # 6: Difference-in-Differences

**Background:** A government training programme for unemployed workers was implemented in Region A starting in January 2023, while Region B did not implement the programme. You have panel data tracking the same individuals in both regions from 2022 and 2023, with the following average earnings (in £1000s):

| Region      | 2022 | 2023 | Sample Size (n) |
|-------------|------|------|-----------------|
| A (Treated) | 18.5 | 24.2 | 250             |
| B (Control) | 20.1 | 22.8 | 300             |

#### 1. DiD Estimation and Identification

- Calculate the canonical  $2 \times 2$  difference-in-differences estimate of the average treatment effect on the treated (ATT). Show your working and interpret the result in the context of the training programme.
- State the three key assumptions (SUTVA, No-Anticipation, and Parallel Trends) required for the DiD estimator to identify the ATT. Explain each assumption in the context of this application.
- Explain intuitively why the parallel trends assumption is necessary. What would happen to the DiD estimate if this assumption were violated?

#### 2. Potential Outcomes and Missing Data

Consider the potential outcomes framework where  $Y_{i,t}(g)$  denotes the earnings individual  $i$  would have in period  $t$  if they belonged to group  $g \in \{A, B\}$ .

- Under SUTVA and No-Anticipation, show which potential outcomes are observed for individuals in each region and time period. Present your answer in a table similar to the one on page 11 of the lecture notes.

- (b) Write down the formal expression for the selection bias that would arise from a simple comparison of post-treatment outcomes:  $\mathbb{E}[Y_{i,t=2023}|G_i = A] - \mathbb{E}[Y_{i,t=2023}|G_i = B]$ . Explain why this does not generally equal the ATT.
- (c) Using the Parallel Trends Assumption, show algebraically how the unobserved counterfactual  $\mathbb{E}[Y_{i,t=2023}(\infty)|G_i = A]$  can be imputed from observed data.

### 3. Regression Specification and Interpretation

Suppose you estimate the following two-way fixed effects (TWFE) regression:

$$Y_{i,t} = \alpha_0 + \gamma_0 \mathbf{1}\{G_i = A\} + \lambda_0 \mathbf{1}\{T_i = 2023\} + \beta_0^{TWFE} (\mathbf{1}\{G_i = A\} \cdot \mathbf{1}\{T_i = 2023\}) + \varepsilon_{i,t}$$

- (a) Interpret each parameter ( $\alpha_0$ ,  $\gamma_0$ ,  $\lambda_0$ ,  $\beta_0^{TWFE}$ ) in terms of conditional expectations of observed outcomes.
- (b) Show that under SUTVA, No-Anticipation, and Parallel Trends,  $\beta_0^{TWFE}$  equals the ATT.
- (c) Explain why  $\beta_0^{TWFE}$  equals the "DiD by hand" estimator you calculated in question 1(a).